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COS 102 — Past Questions & Materials

by Renance

Course: COS 102

Institution: Federal University of Technology, Akure

Sources in this pack: 1

Includes past questions and lecture notes

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BASIC Language – 60 Objective Questions

1–10: Introduction to BASIC

1. BASIC stands for:
 - A. Binary Arithmetic Simple Instruction Code
 - B. Beginner's All-purpose Symbolic Instruction Code ✓
 - C. Basic Algorithm Syntax Instruction Code
 - D. Binary and ASCII System Instruction Code
2. BASIC was developed in the:
 - A. 1960s ✓
 - B. 1970s
 - C. 1980s
 - D. 1950s
3. Which of these is an advantage of BASIC?
 - A. Complex syntax
 - B. Requires advanced mathematics
 - C. Easy to learn ✓
 - D. Only for scientific use
4. BASIC is a:
 - A. Low-level language
 - B. High-level language ✓
 - C. Machine language
 - D. Assembly language
5. The first step in writing a BASIC program is:
 - A. Debugging
 - B. Compiling
 - C. Writing source code ✓
 - D. Saving the program
6. BASIC programs are executed:
 - A. Only in binary form
 - B. Line by line ✓
 - C. After full compilation only
 - D. Without an interpreter
7. Which device is required to run BASIC programs?
 - A. Interpreter ✓
 - B. Scanner
 - C. Compiler only
 - D. Printer
8. BASIC is best described as:
 - A. Procedural ✓
 - B. Object-oriented
 - C. Assembly-based
 - D. Platform-specific

9. BASIC commands are usually written in:
 - A. Uppercase ✓
 - B. Lowercase
 - C. Mixed case only
 - D. Binary
 10. The main purpose of BASIC language is:
 - A. Web design
 - B. Programming for beginners ✓
 - C. Operating system development
 - D. Cryptography
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11–20: BASIC Commands & Syntax

11. The BASIC command to display output on screen is:
 - A. INPUT
 - B. PRINT ✓
 - C. REM
 - D. LET
12. Which BASIC statement is used for comments?
 - A. COM
 - B. REM ✓
 - C. PRINT
 - D. COMMENT
13. The INPUT command is used to:
 - A. Show results
 - B. Receive data from the user ✓
 - C. Stop the program
 - D. Save a file
14. LET in BASIC is used to:
 - A. Assign a value to a variable ✓
 - B. Print output
 - C. Repeat instructions
 - D. Terminate program
15. What does the command `END` do?
 - A. Restarts the program
 - B. Ends program execution ✓
 - C. Deletes variables
 - D. Saves file
16. In BASIC, numbers can be:
 - A. Only integers
 - B. Integers or real numbers ✓
 - C. Only decimal values
 - D. Only binary numbers

17. String variables in BASIC end with:
- A. \$ ✓
 - B. #
 - C. %
 - D. &
18. The statement `PRINT 5+3` will display:
- A. 8 ✓
 - B. 53
 - C. 5+3
 - D. Error
19. The `IF...THEN` statement is used for:
- A. Looping
 - B. Conditional execution ✓
 - C. Data storage
 - D. Program termination
20. The BASIC statement to repeat a block of code is:
- A. `FOR...NEXT` ✓
 - B. `INPUT...PRINT`
 - C. `IF...THEN`
 - D. `REM...PRINT`
-

21–30: BASIC Programming Structure

21. Which keyword initializes a loop in BASIC?
- A. `FOR` ✓
 - B. `LOOP`
 - C. `NEXT`
 - D. `DO`
22. In BASIC, a loop ends with:
- A. `STOP`
 - B. `NEXT` ✓
 - C. `EXIT`
 - D. `BREAK`
23. The BASIC function to calculate square root is:
- A. `SQR` ✓
 - B. `ROOT`
 - C. `SQRT`
 - D. `POW`
24. Which operator is used for exponentiation in BASIC?
- A. `^` ✓
 - B. `**`
 - C. `*`
 - D. `EXP`

25. BASIC uses which symbol for multiplication?
- A. x
 - B. * ✓
 - C. ×
 - D. X
26. Which BASIC function returns absolute value?
- A. ABS ✓
 - B. MOD
 - C. VAL
 - D. FIX
27. BASIC arrays start their index at:
- A. 0 ✓
 - B. 1
 - C. -1
 - D. Any number
28. The BASIC function to convert a string to a number is:
- A. VAL ✓
 - B. STR\$
 - C. NUM
 - D. CONV
29. In BASIC, RND function generates:
- A. Random numbers ✓
 - B. Square roots
 - C. Factorials
 - D. Prime numbers
30. BASIC treats spaces in code as:
- A. Significant
 - B. Ignored ✓
 - C. Errors
 - D. Variables
-

31–40: Advanced BASIC Features

31. The GOTO statement is used for:
- A. Looping
 - B. Unconditional branching ✓
 - C. Ending program
 - D. Declaring variables
32. Which statement is used for subroutines in BASIC?
- A. DEF
 - B. GOSUB ✓
 - C. CALL
 - D. PROC

33. The BASIC command `RETURN` is used to:
- A. Exit program
 - B. Return from subroutine ✓
 - C. Restart program
 - D. Print variable
34. The keyword used to define a function in BASIC is:
- A. `DEF` ✓
 - B. `FUNC`
 - C. `SUB`
 - D. `METHOD`
35. BASIC statements are usually numbered for:
- A. Execution order ✓
 - B. Variable storage
 - C. Debugging only
 - D. Aesthetic purposes
36. The BASIC statement to pause program execution until a key is pressed is:
- A. `STOP`
 - B. `INPUT`
 - C. `INKEY$` ✓
 - D. `WAIT`
37. Which operator is used for string concatenation in BASIC?
- A. `&` ✓
 - B. `+`
 - C. `#`
 - D. `*`
38. BASIC keywords:
- A. Can be used as variable names
 - B. Cannot be used as variable names ✓
 - C. Must be in lowercase
 - D. Are optional
39. In BASIC, logical AND is represented by:
- A. `AND` ✓
 - B. `&`
 - C. `&&`
 - D. `+`
40. The BASIC command to clear variables is:
- A. `CLEAR` ✓
 - B. `ERASE`
 - C. `RESET`
 - D. `CLS`
-

41–50: Input/Output and File Handling

41. The command to clear the screen in BASIC is:
A. CLS ✓
B. CLEAR
C. RESET
D. ERASE
42. Which statement is used to open a file in BASIC?
A. FILE
B. OPEN ✓
C. INPUT
D. SAVE
43. To write data to a file in BASIC, use:
A. PRINT # ✓
B. WRITE
C. OUTFILE
D. SAVE
44. To read data from a file in BASIC, use:
A. INPUT # ✓
B. READ
C. LOAD
D. GET
45. The BASIC command to close an open file is:
A. END FILE
B. CLOSE ✓
C. STOP FILE
D. EXIT
46. The BASIC function to get the length of a string is:
A. LEN ✓
B. LENGTH
C. SIZE
D. COUNT
47. Which BASIC command deletes a file?
A. KILL ✓
B. ERASE
C. REMOVE
D. DELETE
48. The BASIC function to convert number to string is:
A. STR\$ ✓
B. NUM\$
C. TOSTR
D. CHR
49. The BASIC command to read data from DATA statements is:
A. READ ✓
B. INPUT
C. LOAD
D. GET

50. The BASIC statement to skip to the next DATA item is:
- A. RESTORE ✓
 - B. NEXT
 - C. CONTINUE
 - D. JUMP
-

51–60: Debugging & Error Handling

51. Which BASIC command is used to stop program execution for debugging?
- A. STOP ✓
 - B. END
 - C. PAUSE
 - D. BREAK
52. BASIC displays syntax errors:
- A. Before running program ✓
 - B. After complete execution
 - C. Randomly
 - D. Never
53. Runtime errors occur:
- A. Before program runs
 - B. While program is running ✓
 - C. Only in compiled languages
 - D. Never in BASIC
54. The BASIC command to trace program execution is:
- A. TRACE ✓
 - B. TRACK
 - C. DEBUG
 - D. STEP
55. Logical errors in BASIC:
- A. Prevent execution
 - B. Produce wrong results ✓
 - C. Crash the system
 - D. Cause compilation failure
56. The BASIC command to remove all variables from memory is:
- A. CLEAR ✓
 - B. ERASE
 - C. DEL
 - D. CLS
57. Which BASIC statement can be used for error trapping?
- A. ON ERROR GOTO ✓
 - B. TRY...CATCH
 - C. IF ERROR THEN
 - D. RESUME

58. In BASIC, the command `LIST` is used to:
- A. Display the program code ✓
 - B. Show variables
 - C. Save program
 - D. Run program
59. Which command runs the BASIC program?
- A. `START`
 - B. `RUN` ✓
 - C. `PLAY`
 - D. `EXECUTE`
60. The BASIC statement `CONT` is used to:
- A. Continue execution after a stop ✓
 - B. Restart program
 - C. Exit program
 - D. Repeat a loop

Here are 60 multiple-choice questions (MCQs) for undergraduate Computer Science students based on the uploaded *A Textbook of BASIC Language* content you provided.

Chapter 1: Introduction to BASIC Programming

1. Which of the following is NOT a programming language?
(a) COBOL (b) Pascal (c) CorelDraw (d) BASIC
 2. A programming language coded using mnemonics is called:
(a) Assembly language (b) Machine language (c) High-level language (d) Mobile language
 3. The term BASIC stands for:
(a) Binary Arithmetic Symbolic Instruction Code (b) Beginner's All-purpose Symbolic Instruction Code
(c) Basic Application Syntax in Computing (d) Binary Application System for Integrated Computing
 4. Which programming language was originally designed for teaching programming to beginners?
(a) FORTRAN (b) BASIC (c) COBOL (d) PASCAL
 5. Machine language consists of:
(a) Mnemonics (b) Natural language words (c) Binary digits (d) Flowcharts
 6. A high-level language must be:
(a) Written in binary (b) Translated into machine code (c) Understood directly by hardware (d) Written with mnemonics
 7. Which of the following is an example of an object-oriented programming language?
(a) BASIC (b) C++ (c) Assembly (d) FORTRAN
 8. Which language is commonly used in business data processing?
(a) BASIC (b) COBOL (c) PASCAL (d) FORTRAN
 9. In object-oriented languages, data and code are grouped into:
(a) Functions (b) Objects (c) Procedures (d) Records
 10. Logic programming languages are often used in:
(a) Artificial intelligence (b) System drivers (c) Game design (d) File processing
-

Chapter 2: Programming Process and Design

11. The first stage of program development is:
(a) Coding (b) Testing (c) Problem analysis (d) Debugging
12. Which of the following represents a narrative description of program steps?
(a) Flowchart (b) Source code (c) Pseudo code (d) Syntax
13. In a flowchart, the diamond shape represents:
(a) Process (b) Input/Output (c) Decision (d) Start/Stop
14. Which flowchart symbol represents output operations?
(a) Rectangle (b) Parallelogram (c) Oval (d) Pentagon

15. A step-by-step procedure for solving a problem in a finite number of steps is called:
(a) Flowchart (b) Algorithm (c) Program (d) Syntax
 16. Which of these is NOT a rule for flowcharting?
(a) Use standard symbols (b) Cross flow lines when needed (c) Maintain consistent spacing (d) Draw with pencil
 17. Which programming design tool uses English-like statements?
(a) Flowchart (b) Algorithm (c) Pseudo code (d) Logic table
 18. The process of removing errors in a program is called:
(a) Compiling (b) Debugging (c) Executing (d) Coding
 19. In pseudo code, variables are usually:
(a) Written in binary (b) Written in natural language (c) Written in a mix of English and programming syntax (d) Written as mnemonics
 20. Flowcharts are standardized by:
(a) IEEE (b) ANSI (c) ISO (d) W3C
-

Chapter 3: BASIC Program Development Interface

21. The QBASIC menu used to execute a program is:
(a) File (b) Run (c) Edit (d) View
 22. The command to save a program in QBASIC is found under:
(a) File (b) Edit (c) Run (d) Option
 23. Which QBASIC menu contains the Cut, Copy, and Paste commands?
(a) File (b) Edit (c) Run (d) Option
 24. The shortcut key to run a program in QBASIC is:
(a) F1 (b) F5 (c) F7 (d) F9
 25. The Immediate Mode in QBASIC is used to:
(a) Execute code without running the main program (b) Save programs quickly (c) Debug programs (d) Edit files
 26. The status bar in QBASIC shows:
(a) File size (b) Cursor position (c) Disk capacity (d) Compilation errors only
 27. To open a saved program in QBASIC, you use:
(a) Edit → Open (b) File → Open (c) Run → Open (d) Option → Open
 28. Which QBASIC menu allows you to change display colors?
(a) File (b) Edit (c) Options (d) View
 29. In QBASIC, "SUBs" under the View menu displays:
(a) Current subroutines and functions (b) Saved programs (c) Program variables (d) Output
 30. The command to quit QBASIC is:
(a) File → Close (b) File → Exit (c) Run → Stop (d) Edit → Exit
-

Chapter 4: Data, Constants and Variables

31. A variable name ending with \$ represents:
(a) Numeric data (b) String data (c) Integer (d) Boolean
32. Which of the following is NOT a numeric data type in BASIC?
(a) Integer (b) Double precision (c) Single precision (d) String
33. A value that does not change during program execution is called:
(a) Variable (b) Constant (c) Identifier (d) Keyword
34. Which BASIC reserved word is used to clear the screen?
(a) CLS (b) CLEAR (c) END (d) RESET
35. The BASIC statement to display output is:
(a) INPUT (b) PRINT (c) READ (d) DISPLAY
36. Which of these is NOT an arithmetic operator in BASIC?
(a) + (b) - (c) ** (d) /
37. The BASIC reserved word used to send output to the printer is:
(a) WRITE (b) LPRINT (c) PRINT (d) OUTPUT
38. Which error type results from faulty algorithm or program logic?
(a) Syntax error (b) Runtime error (c) Logical error (d) Execution error
39. The operator ^ in BASIC is used for:
(a) Division (b) Exponentiation (c) Modulus (d) Multiplication
40. In BASIC, reserved words:
(a) Can be used as variable names (b) Have predefined meanings (c) Are optional in all cases (d) Are only used for comments

Chapter 5: Control Structure

41. Relational operators are used for:
(a) Arithmetic calculations (b) Logical comparisons (c) String concatenation (d) Input
42. Which relational operator means “not equal to”?
(a) = (b) <> (c) >= (d) <=
43. The IF...THEN statement is used for:
(a) Looping (b) Conditional branching (c) Printing output (d) Reading input
44. In an IF...THEN...ELSE structure, the ELSE block executes when:
(a) The condition is true (b) The condition is false (c) The program ends (d) The program starts
45. In BASIC, a multi-branch selection can be implemented with:
(a) FOR...NEXT (b) GOTO (c) ELSEIF (d) DIM
46. An unconditional branch transfers control:
(a) Based on a condition (b) Without any condition (c) Only in loops (d) Only with arrays
47. Which symbol represents “less than or equal to” in BASIC?
(a) <= (b) =< (c) => (d) ><
48. A block of statements that runs only if a condition is true is called:
(a) Loop (b) Procedure (c) Conditional block (d) Function

49. In BASIC, CASE is used for:
(a) String formatting (b) Multiple selection (c) File handling (d) Array processing
50. The ELSEIF clause is used:
(a) Inside loops (b) In nested decisions (c) For variable declaration (d) For file reading

Mixed Chapters / Application

51. Which of the following is NOT a valid BASIC variable name?
(a) X1 (b) COUNT (c) NAME\$ (d) END
52. In QBASIC, the file extension for saved programs is typically:
(a) .TXT (b) .BAS (c) .EXE (d) .DOC
53. The BASIC reserved word for entering values during program execution is:
(a) INPUT (b) READ (c) DATA (d) PRINT
54. Which of these BASIC statements is optional in modern versions of BASIC?
(a) CLS (b) END (c) LET (d) INPUT
55. A BASIC statement that stores a value into a variable is called:
(a) Assignment statement (b) Declaration (c) Execution (d) Control statement
56. What is the correct hierarchy of arithmetic operators in BASIC?
(a) Addition → Multiplication → Exponentiation (b) Multiplication → Addition → Exponentiation
(c) Exponentiation → Multiplication/Division → Addition/Subtraction (d) Multiplication/Division → Addition/Subtraction → Exponentiation
57. In BASIC, comments are written using:
(a) REM or ' (b) // (c) # (d) /.../
58. The term "GOTO" in BASIC is used for:
(a) Looping (b) Branching (c) File opening (d) Ending a program
59. Which BASIC error type occurs when dividing a number by zero?
(a) Syntax error (b) Logical error (c) Runtime error (d) Typing error
60. Which is the correct BASIC statement to output the sum of variables A and B?
(a) PRINT "A+B" (b) PRINT A + B (c) PRINT A,B (d) DISPLAY A+B